



HAL 9000 keeps astronauts alive

Scientists are developing a HAL 9000-inspired AI that will help astronauts with murdering them, like the OG. <https://dgit.in/dec18-20-1>



Razor protection plan

Razor is offering 3-year protection plans for their gaming laptops and smartphones under RazorCare Essential. <https://dgit.in/dec18-20-2>

- **Lip sync:** While audio editing software has made it possible to modify speech to be near identical to a target, most fake generating tools have not been able to achieve a high enough level of sophistication to generate speech accurately. Even if you take into account a professional mimicry artist, in most doctored videos today the disparity between lip movements and the actual speech is too evident. You should also listen to a couple of verified videos of the person involved to familiarize yourself with their intonation, pronunciation, etc., and you'll be able to spot the difference every single time.
- **Uncoordinated facial movements:** Keep an eye out for facial movements that you cannot explain or understand. For instance, the face and the mouth might be moving in completely different directions, same goes for the eyes, as well as coordination with the body. Jerky, out of place movements are a result of an unfiltered data source and a purely computational fake generating process with no human intervention.
- **Lighting:** A crucial aspect of all videos, especially outdoor videos, lighting can often be a dead giveaway for a faked video. A simple way to spot outdoor fakes is to plot the location of the sun with respect to the shadows it creates. This isn't as difficult as it sounds, all you'll need to do is take a frame where something questionable is happening and draw lines along the shadows created by all the objects. If they don't meet up at the same point, you're dealing with a fake. Of course, if objects are missing shadows, to begin with, that's a definite fake.

STEP 3: CUTTING EDGE

The threat that professionally made fake videos pose to the world is not lost on anyone, including programmers and researchers. Hence, apart from ways for you to detect these videos on your own, there are tools and projects being worked on that can

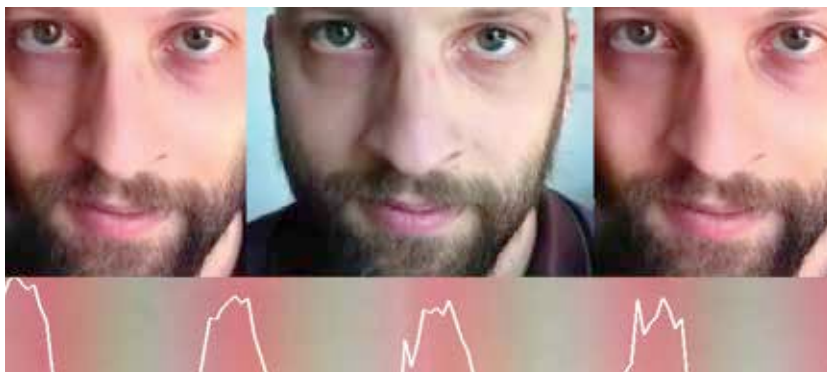


Source: YouTube

The viral video of an eagle snatching a kid was debunked by plotting the sun's location based on the shadows

detect fake videos for you. Some of them have been proven to be efficient while some are still in the testing stages. Some of these methods depend on spotting changes that are not perceptible to human cognition.

A team of researchers from the University of Trento, Trento (Italy) and Dartmouth College came up with a method to differentiate real faces from CGI faces. Essentially, a human face changes colour ever so slightly



The colour change of the human face with every pulse

with every pulse. As blood flows into the face, it turns redder, and the opposite movement turns it greener. Their program detected this phenomenon and could differentiate between face-swapped deepfake videos. Since the faces generated for these videos were based on a database of still images, the necessary variation for the pulse wasn't included. However, Hany Farid, one of the researchers behind this technology, doesn't think it'll take too long for some random

Reddit to figure this out and encode the pulse into deepfakes as well.

STEP 4: LOOK BEYOND TECHNOLOGY FOR AN ANSWER

No, we're not talking about using common sense to detect videos that have been modified using technology. Some of the most effective fake videos aren't really fake at all, just taken out of context.

India currently faces a fake media epidemic, where low-tech fakes circulated via social media and messaging apps have caused more damage than any tech-infused forgeries have. Most of these fakes just have captions that put the video into a false context, and people don't bother to check the authenticity, source or accuracy of translated videos recorded originally in foreign languages. Their effectiveness relies on your tendency to take things at face

value and not bother about verifying the facts, especially if the video suits a particular agenda that you support.

Getting beyond this behaviour is really important. The technology to make better fake videos will catch up and overtake the technology to detect it, and this video 'arms' race between the two sides will keep going on as long as tech keeps progressing. How we react to the next viral video or WhatsApp forward is what will make the real difference... 